

RTL-to-GDS Final Evidence Report

RISC-V FIR accelerator and fine-grained DVFS adaptive clock-gating projects

This report summarizes the verified RTL simulation and OpenROAD Sky130HD physical-design flow completed locally and pushed to GitHub.

Completion Checklist

Stage	RISC-V FIR	DVFS Demo
RTL cleanup/top selection	Done	Done
Self-checking RTL simulation	Done	Done
VCD waveform generation	Done	Done
Sky130HD synthesis	Done	Done
Floorplan	Done	Done
Placement	Done	Done
Clock tree synthesis	Done	Done
Global/detailed routing	Done	Done
Final timing closure	Done	Done
DRC and antenna checks	Done	Done
Final GDS generation	Done	Done
KLayout GDS screenshot	Done	Done
GitHub documentation push	Done	Done

Headline Metrics

Metric	RISC-V FIR Accelerator
Top module	riscv_fir_accelerator_top
Simulation	Self-checking Icarus Verilog tests pass; VCD generated
Minimum clock period / Fmax	9.60 ns / 104.14 MHz
Final design area	7610 um^2
Final utilization	44%
Timing	WNS 0.00 ns, TNS 0.00 ns, setup/hold violations 0
Physical checks	Detailed-route DRC 0, antenna violations 0
Estimated total power	1.30 mW
Worst VDD IR drop	0.102 mV

Metric	DVFS / Adaptive Clock-Gating Demo
Top module	dvfs_demo_top
Simulation	Self-checking smoke test passes; VCD generated
Minimum clock period / Fmax	7.40 ns / 135.16 MHz
Final design area	3193 um^2
Final utilization	36%
Timing	WNS 0.00 ns, TNS 0.00 ns, setup/hold violations 0
Physical checks	Detailed-route DRC 0, antenna violations 0
Estimated total power	0.55 mW
Worst VDD IR drop	0.092 mV

Simulation Evidence

The FIR testbench shows reset release, valid input samples, output valid assertions, and the expected accumulated FIR result sequence.

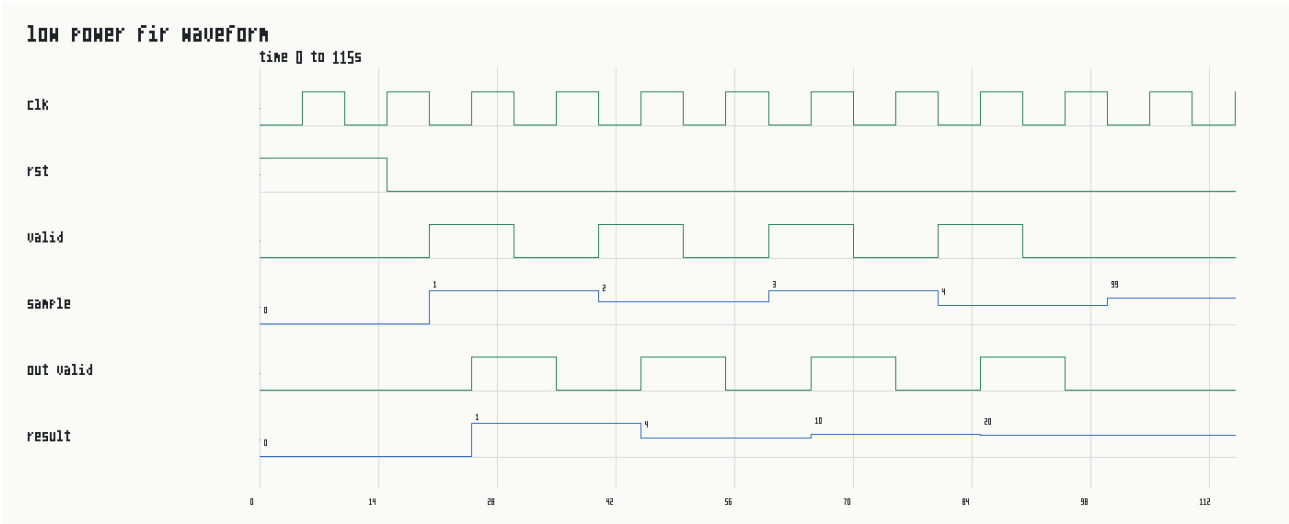


Figure 1. Low-power FIR waveform.

The DVFS smoke test shows PMU writes, requested voltage movement from nominal to high, regulator unstable/stable behavior, sleep assertion, wake assertion, and gated-clock recovery.

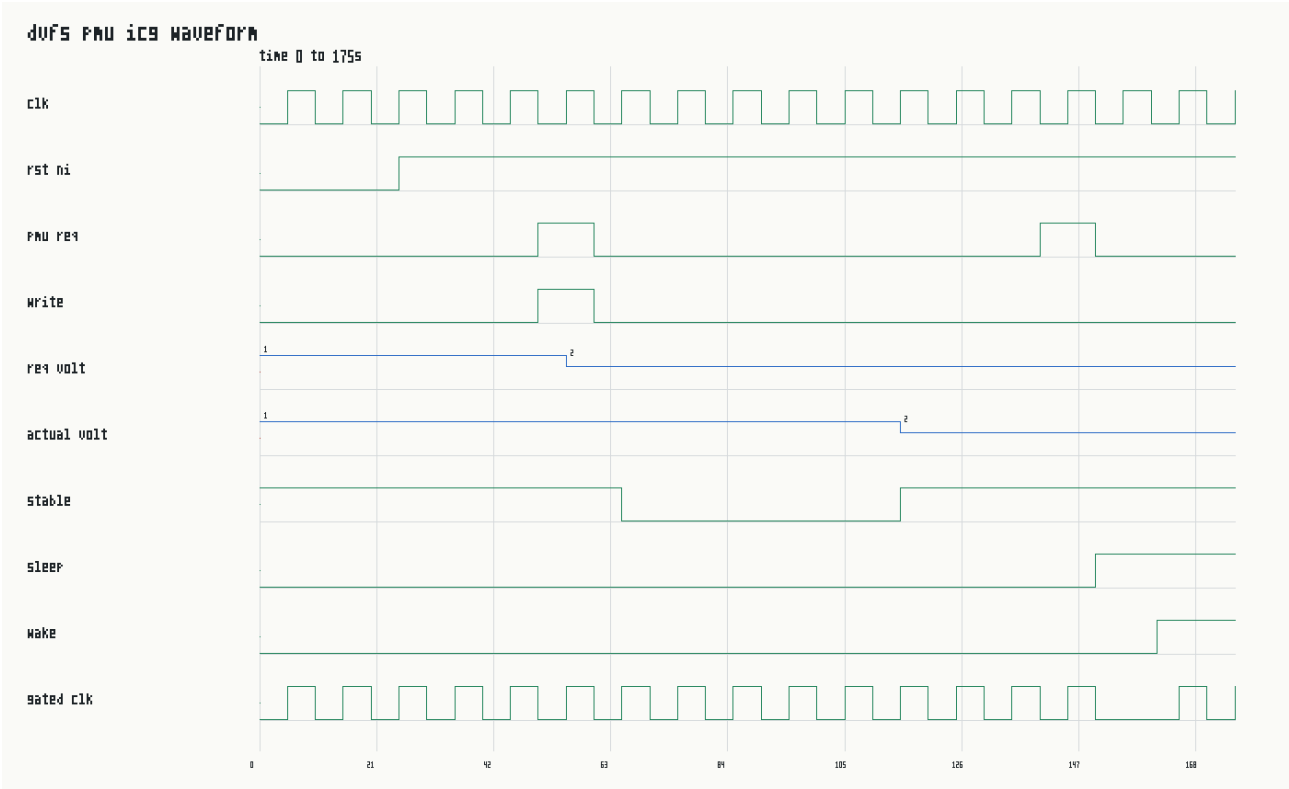


Figure 2. DVFS PMU and clock-gating waveform.

GDS Layout Evidence

Both designs completed final GDS generation through OpenROAD-flow-scripts using the Sky130HD platform. KLayout was used to render the final layouts.

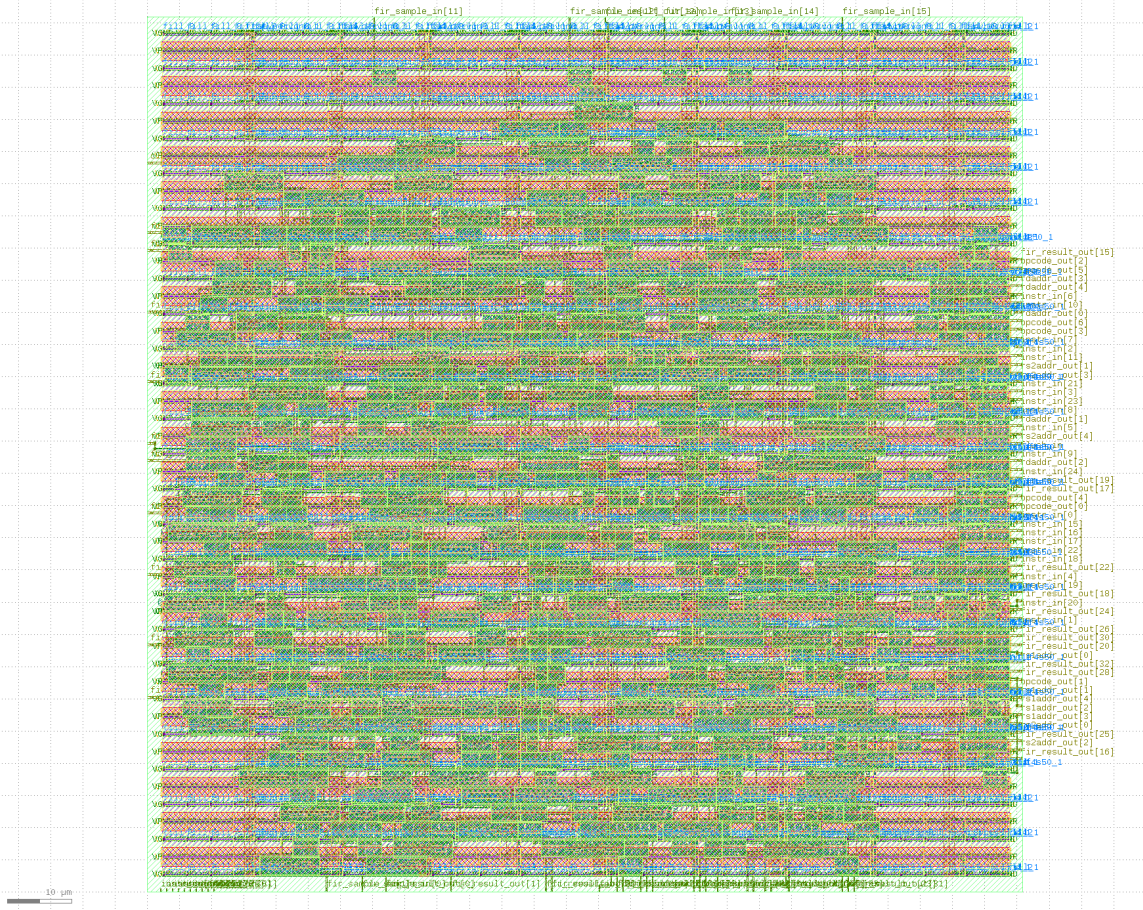


Figure 3. RISC-V FIR accelerator final GDS layout.

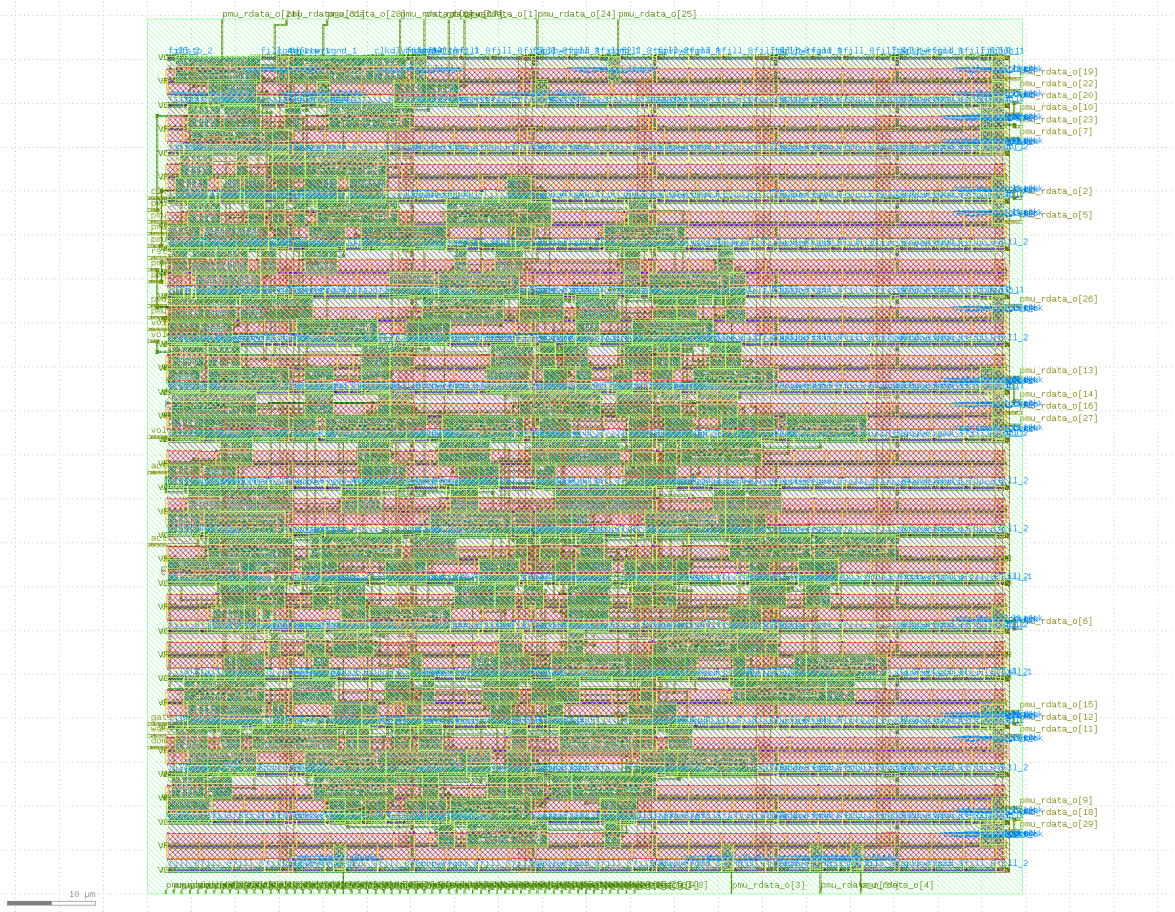


Figure 4. DVFS demo top final GDS layout.

Remaining Engineering Note

The DVFS repository's full Ibex-based processor integration remains a future milestone because the external lowRISC Ibex RTL is not vendored under `rtl/core/ibex`. The current complete RTL-to-GDS proof uses `dvfs_demo_top`, which exercises the project-owned PMU, regulator, clock-gating, and CDC logic.